

Assumption :

$$\text{Link 1 } (L_1) = 10 \text{ cm}$$

$$\text{Link 2 } (L_2) = 7 \text{ cm}$$

$$\text{Joint 1 } (\theta_1) = 30^\circ$$

$$\text{Joint 2 } (\theta_2) = 45^\circ \leftarrow L_2 \text{ relative to } L_1$$

Find (x, y) coordinates of end effector

Soln :

$$\begin{aligned} x &= L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ &= L_1 \cos(30^\circ) + L_2 \cos(30^\circ + 45^\circ) \\ &\approx 10(0.866) + 7(0.259) \\ &\approx 8.66 + 1.813 \\ &\approx 10.473 \text{ cm} \end{aligned}$$

$$\begin{aligned} y &= L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \\ &= L_1 \sin(30^\circ) + L_2 \sin(30^\circ + 45^\circ) \\ &\approx 10(0.5) + 7(0.966) \\ &\approx 5 + 6.762 \\ &\approx 11.762 \text{ cm} \end{aligned}$$

hence for $L_1 = 10$, $L_2 = 7$ & $\theta_1 = 30^\circ$, $\theta_2 = 45^\circ$

the position of the end effector is $(10.473, 11.762)$
(and Joint 1 position will be $(8.66, 5.00)$)